CODEFORCES
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PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

D. Fast Matrix Drift

time limit per test: 1 second
memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given a 2×2 integer matrix A and an integer exponent X $(1 \le X \le 10^9)$.

Your task is to compute the matrix A^X, where all intermediate operations (additions and multiplications during matrix multiplication) are performed modulo $10^9 + 7$.

Formally, for two matrices:

$$B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}, \quad C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}.$$

Their product $D = B \times C$ is defined as: $D_{ij} = (b_{i1} \cdot c_{1j} + b_{i2} \cdot c_{2j}) \bmod (10^9 + 7)$.

Input

The first line contains a single integer T $(1 \le T \le 10^5)$ — the number of test cases.

The first line of each test case contains four integers a_{11}, a_{12}, a_{21}, and a_{22} where $0 \le a_{ij} \le 10^9$ — the elements of the matrix A. The second line contains an integer X where $1 \le X \le 10^9$.

Output

For each test case, print two lines, each containing two integers: the resulting 2×2 matrix A^X. Each element should be printed modulo $10^9 + 7$.

Examples

input Copy

```
2
1 1 1 0
5
2 0 0 2
3
```

output Copy

```
8 5
5 3
8 0
0 8
```

input Copy

```
3
1 1 1 1
10
1 2 3 4
12
1 2 1 2
5
```

output Copy

CSE221 Assignment 03 Fall 2026
Contest is running
6 days
Contestant

→ Submit?

Language: Java 21 64bit
Choose file: Choose File No file chosen
Success!
CLOUDFLARE
Privacy • Help
Submit

```
512 512
512 512
138067399 201223170
301834755 439902154
81 162
81 162
```

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